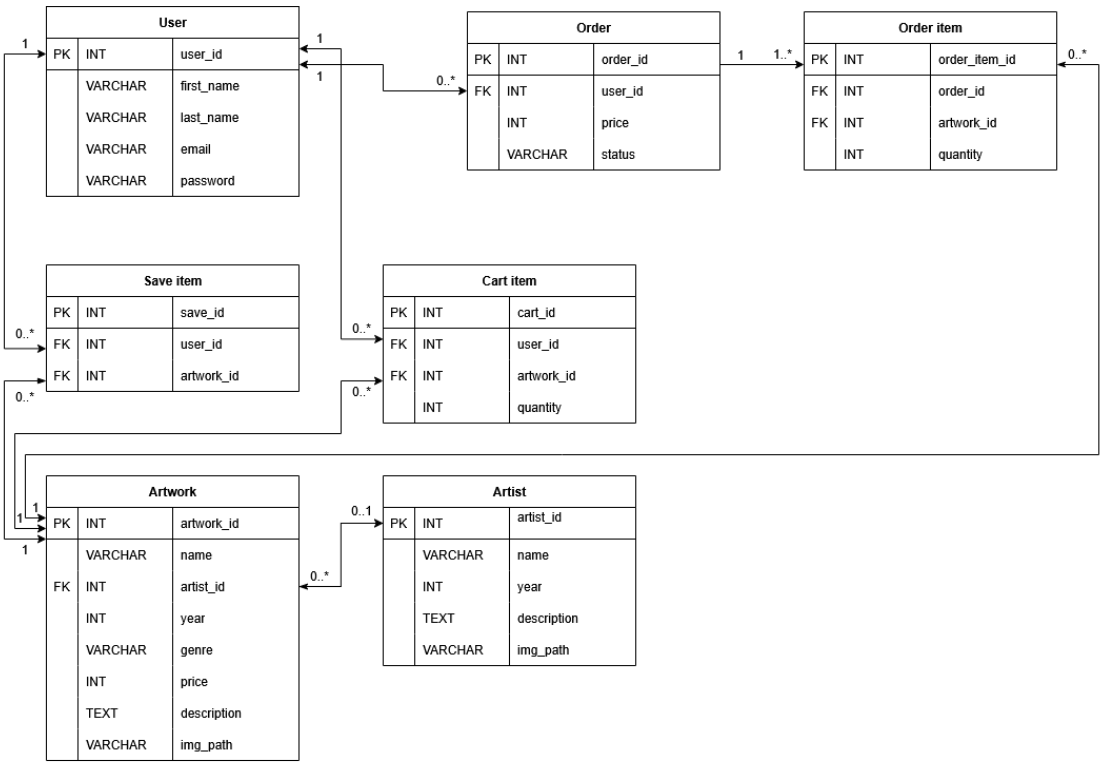
Daniil Stroievskyi, Dmytro Shapovalov

WTECH Project (1 Phase)

1. Assignment

Create a web-based e-commerce application for an online art gallery (DSgallery).

2. Physical data model



The diagram contains 7 entities with the following structure and relationships:

User stores registered customers with their basic information (user_id, first_name, last_name, email, password). It is the central entity, one user can have zero or more orders, zero or more saved items, and zero or more cart items.

Order represents a completed purchase (order_id, user_id, price, status). Each order belongs to exactly one user. One order contains one or more Order Items.

Order Item is a line item within an order (order_item_id, order_id, artwork_id, quantity). It links an order to a specific artwork and records the quantity purchased.

Cart Item represents the user's active shopping cart (cart_id, user_id, artwork_id, quantity). One user can have zero or more cart items, each pointing to one artwork.

Save Item represents the user's bookmarks (save_id, user_id, artwork_id). One user can save zero or more artworks.

Artwork is the core product entity (artwork_id, name, artist_id, year, genre, price, description, img_path). Each artwork belongs to one artist. Artworks are referenced by Cart Items, Save Items, and Order Items.

Artist stores information about the painter (artist_id, name, year, description, img_path). One artist can have zero or more artworks associated with them.

Relationships:

- User → Order: 1 to 0..*
- User → Cart Item: 1 to 0..*
- User → Save Item: 1 to 0..*
- Order → Order Item: 1 to 1..*
- Artist → Artwork: 1 to 0..*
- Artwork → Cart Item / Save Item / Order Item: referenced by all three via artwork_id FK

3. Design decisions

External libraries:
Bootstrap 5.3 (CDN) - used for responsive grid, form controls, buttons, pagination, and navbar (recommended by the course).

4. Programming environment

HTML5, CSS3 (Framework – Bootstrap 5.3.0 (CDN)), Javascript (not yet implemented).

5. Use case implementation

Quantity change - the cart page (cart.html) contains a quantity control UI with − / + buttons and a number input per item.

Login - the login page (login.html) has email and password fields. The "Log in" button redirects the user to home.html with no validation or authentication logic.

Search - a search input is present in the navbar across all pages.

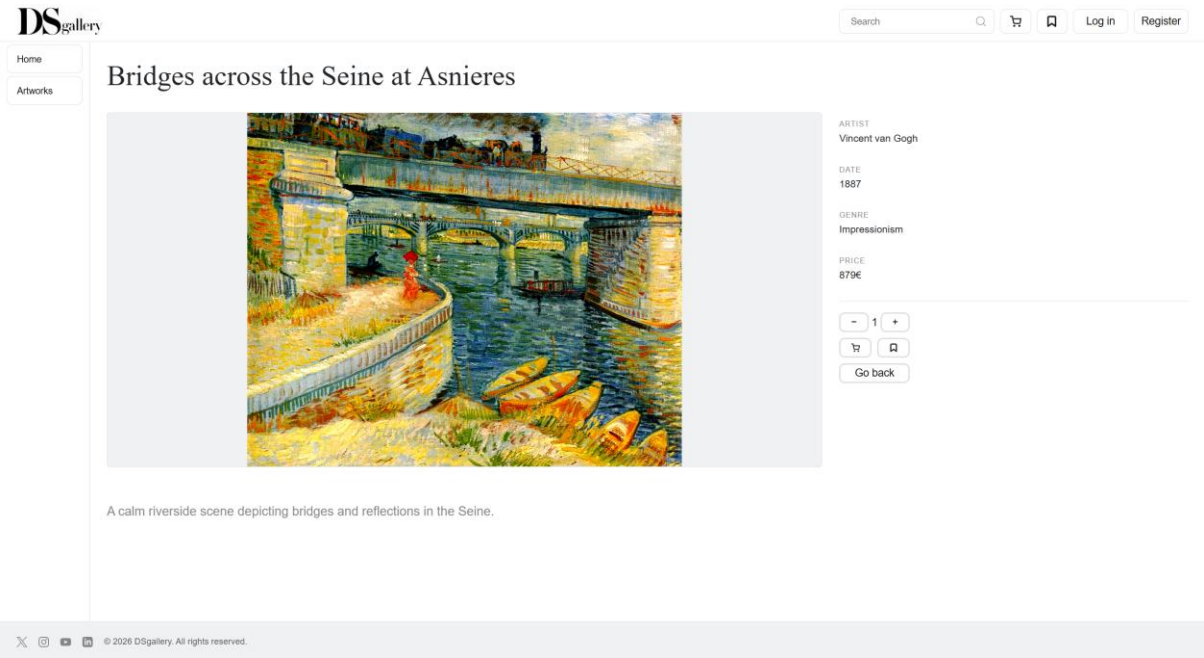
Add to cart - each artwork tile has a bag icon button styled to indicate cart/saved state.

Pagination - bootstrap pagination UI is present on the artworks page.

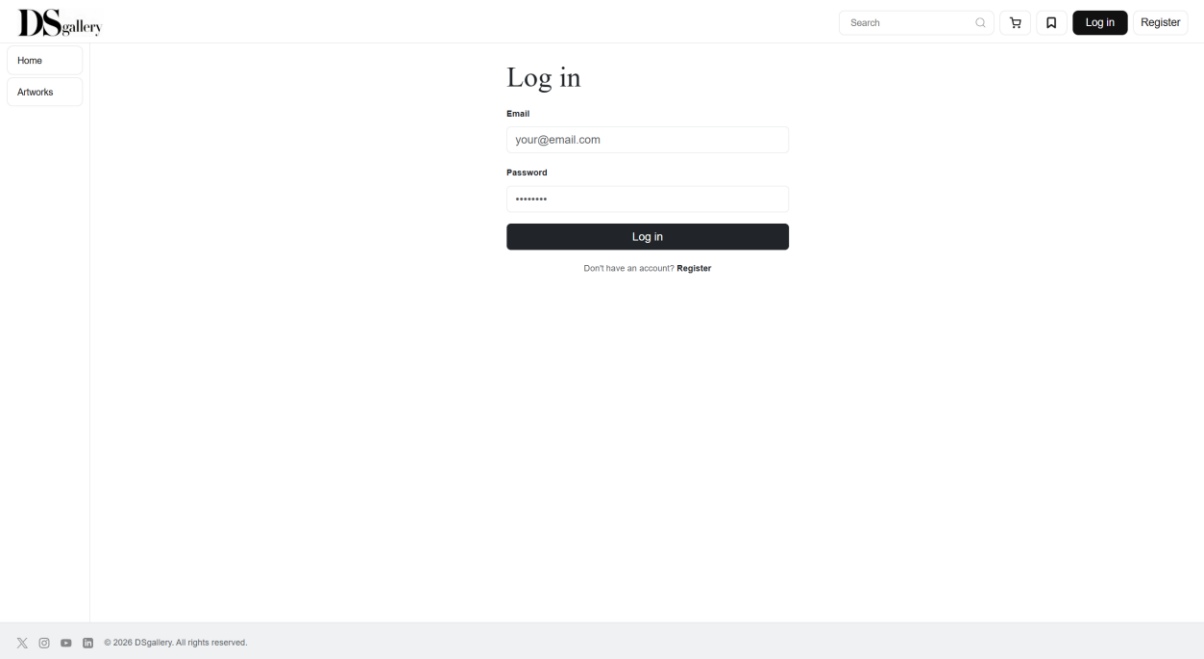
Basic filtering - the artworks page has a filter panel with price range slider, year range slider, and genre checkboxes.

6. Screenshots

Product detail:



Log in:



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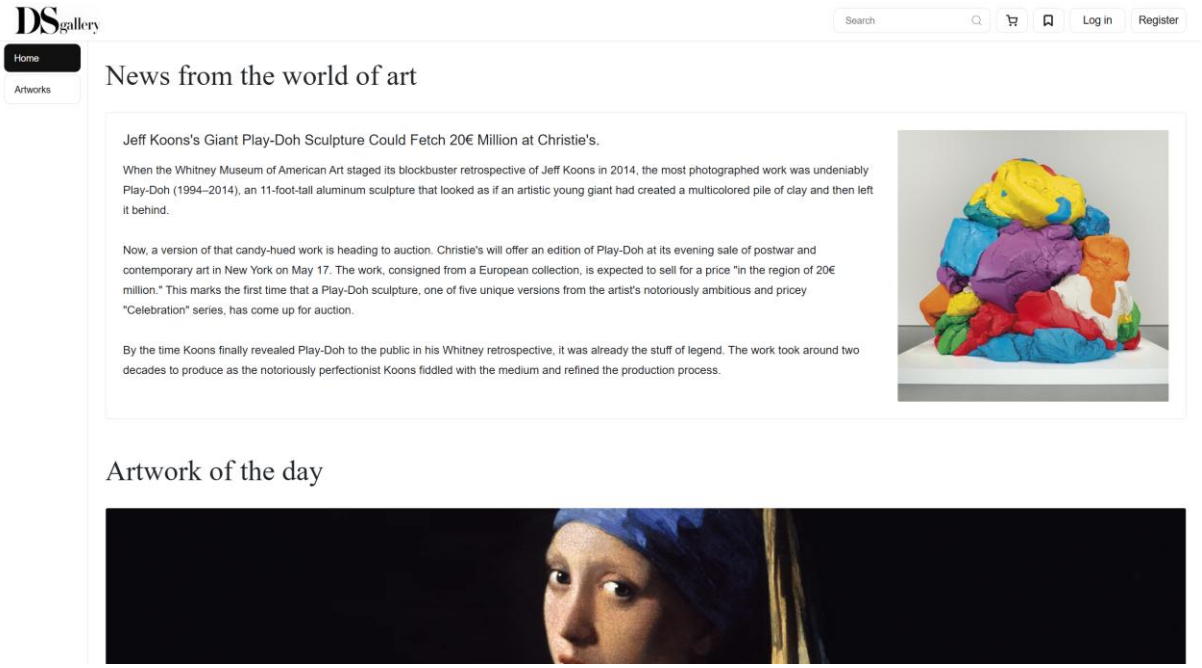
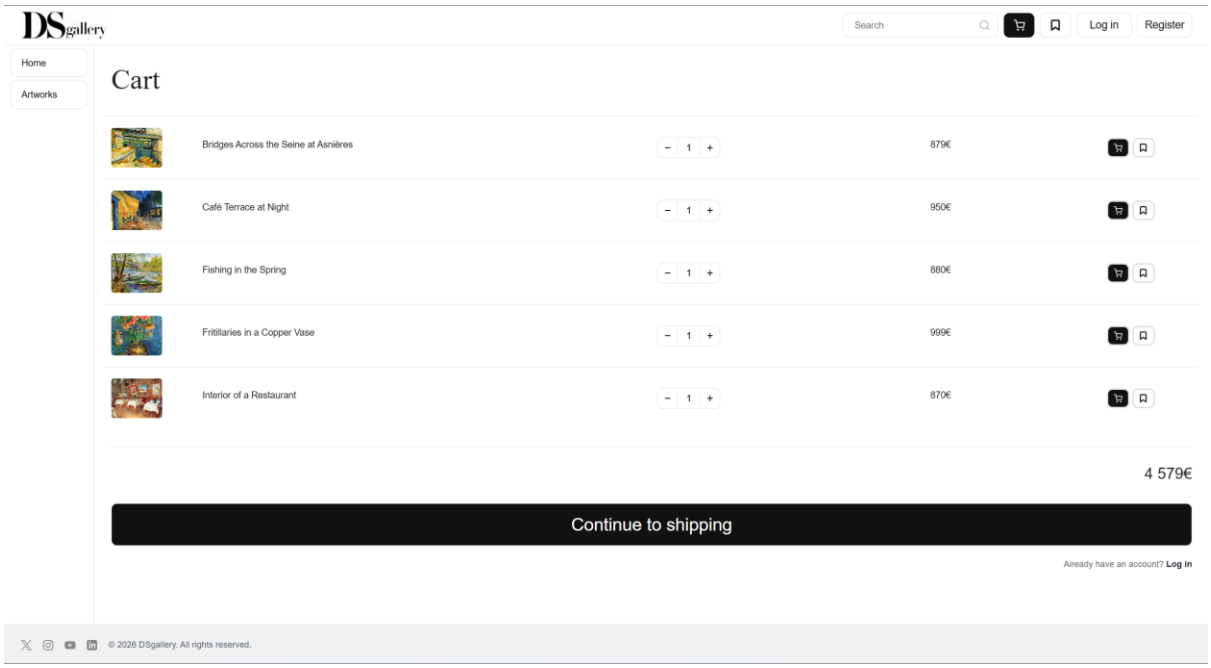
Homepage:



Cart: